

# Usage

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## Basic Usage

### 1. Open the Tool

Select **dennokoworks** > **Gradation Baker** from the menu to open the window.

Check the "Enable Tool" checkbox at the top right of the window to activate the UI. When unchecked, all operations are locked.

### 2. Add Meshes

Drag and drop SkinnedMeshRenderer, MeshRenderer, or any GameObject with a Renderer component from the Hierarchy into the drop area in the Target Meshes section. You can add multiple meshes.

Expand the "►" icon to the left of each mesh to access detailed settings:

- **UV Channel:** Target UV channel for baking (UV0–UV3)
- **Mask Texture:** R channel of texture used as mask (optional)
- **Use Vertex Color:** R channel of vertex color used as mask
- **Invert Mask:** Invert mask application
- **Split by Material:** Outputs separate textures for each sub-mesh (material), useful when UVs overlap.

Click "Clear All" to remove all meshes from the list.

### 3. Set the Gradient

Configure colors in the "Gradient" section. Click on the gradient editor to modify colors.

### 4. Adjust Gradient Range

Use the box handles in the Scene View to adjust the gradient range and direction.

You can also enter values directly in the window fields:

- **Shape:** Choose between **Linear** (directional gradient) and **Spherical** (radial gradient from center).
- **Center:** Center position of the box (use the "Reset" button to automatically fit to mesh bounds)
- **Rotation:** Gradient direction (use the "Reset" button to reset to default)
- **Height** (Linear) / **Size** (Spherical): Scale of the gradient range
- **Rotation Handle:** Change gradient direction
- **Position Handle:** Move the entire box or sphere center
- **Red/Green Cones:** Adjust dimensions (Height for Linear, Width/Height/Depth for Spherical). The gradient starts at 0% in the center/bottom and ends at 100% at the surface/top.

### 5. Bake & Save

Click the "Bake & Save" button to generate and save the texture. The UV channel and mask settings configured for each mesh will be applied.

## Advanced Features

### Work Mesh

When using non-destructive scale components like ModularAvatar, coordinate mismatches can occur. Create a work mesh copy to avoid this issue.

- When no work mesh exists, a "**Create Work Mesh**" button is shown. Click it to create working copies of all meshes.
- When work meshes exist, the button changes to "**Delete Work Mesh**" to remove them.
- Gradient range is automatically adjusted to fit the mesh when work meshes are created.

### Gradient Save/Load

Click "Save" to save the gradient as a texture, and "Load" to restore it. Default save location is [Assets/GeneratedGradation/gradation/](#).

### Mirror

Enable the toggle in the "Mirror" section to access the following settings:

- **Mirror Axis:** Mirror the gradient symmetrically along X, Y, or Z axis. A cyan box indicates the mirror position.
- **Overlap Blend:** Choose how overlapping mirrored areas are blended.
  - **Brighter (Max):** Blend by taking the brighter color.
  - **Darker (Min):** Blend by taking the darker color.

### Preview

Use the "Blend Mode" dropdown in the "Preview" section to switch how the gradient overlay is composited in the Scene View.

**Note:** The preview overlay is simplified. The generated texture colors correspond to an opacity of 1 (fully opaque).

### Dithering

This feature reduces color banding (visible steps or stripes in the gradient) caused by limited color precision during texture output and previews.

- **Dithering Mode (Dithering):**
  - **None:** No dithering is applied.
  - **Interleaved Gradient Noise (IGN):** Applies high-frequency noise in pixel space. It provides a sharp, structured noise that quickly and cleanly reduces banding (default).
  - **Triangular Noise (TPDF):** Applies white noise with a triangular probability density function (TPDF). It yields a more random, uniform noise distribution.
- **Dither Intensity:**
  - Adjusts the strength of the noise (0.0 to 2.0). Higher values make the noise more noticeable but reduce color banding more effectively.

**Tip:** Dithering automatically applies an inverse correction for sRGB perceptual space (square root scaling) based on human visual characteristics. This minimizes unwanted noise scaling in darker regions while effectively eliminating banding.

Output Settings

| Setting                             | Description                                                                                                                  |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Resolution                          | Output texture resolution                                                                                                    |
| Output to material's texture folder | Save to the same folder as the mesh's material textures (ON by default)                                                      |
| Save Path                           | Save destination when the above is OFF (use "... " button to browse)                                                         |
| Background                          | Choose from Transparent / White / Black                                                                                      |
| Edge Padding                        | Expands UV island edges outwards (0–16px) to prevent seam artifacts during sampling                                          |
| Dithering                           | Applies a subtle noise to smooth out color gradients and prevent color banding (IGN or TPDF, default: IGN at 1.5 intensity). |

**Tip:** If you notice "seams" due to MIP mapping or texture sampling, setting Edge Padding to 4px or higher is recommended.

**Note:** After saving, the first saved texture will automatically be selected and revealed in the Project tab.

Language Toggle

Use the "Enable English Mode" checkbox at the top right of the window to switch between English and Japanese.